AE4ASM505 Non-linear Modelling (using F.E.M.)

Homework Assignment 2

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1 Problem A: Contact problem between truss and beam elements

The structural system consists of a vertical truss element (Nodes 3-4) whose lower node (Node 3) is in contact with the central node (Node 2) of a two-element Euler-Bernoulli beam mesh (Nodes 1-2-5). Nodes 1 and 5 are fully clamped (all degrees of freedom equal to zero). The truss nodes are constrained to move only vertically. A downward displacement $\Delta = 0.01$ m is prescribed at Node 4. Contact between Node 3 and Node 2 is enforced via the penalty method. All elements follow the Total Lagrangian formulation from Lecture 3.

All elements share the same material and cross-sectional properties:

- Initial element length: $L = 1$ m
- Square cross-section of side $b = 0.03$ m, so:

$$A_0 = b^2 = (0.03)^2 = 9 \times 10^{-4}\ \text{m}^2, \qquad I_z = \frac{b^4}{12} = 6.75 \times 10^{-8}\ \text{m}^4 \tag{1.1}$$

- Young's modulus: $E = 80$ GPa

The key derived stiffness quantities are:

$$\frac{EA_0}{L} = 7.2 \times 10^7\ \text{N/m}, \qquad K_{\text{pen}} = 10\,\frac{EA_0}{L} = 7.2 \times 10^8\ \text{N/m} \tag{1.2}$$

$$\frac{24\,EI_z}{L^3} = 1.296 \times 10^5\ \text{N/m}, \qquad \frac{8\,EI_z}{L} = 4.32 \times 10^4\ \text{N}\,\text{m/rad} \tag{1.3}$$

1.1 Beam elements (Nodes 1-2 and 2-5)

Each Euler-Bernoulli beam element has DOF vector $\mathbf{a}_{\text{b}}^T = [w_1,\, \theta_1,\, w_2,\, \theta_2]$ and Total Lagrangian tangent stiffness $\mathbf{K}_{\text{b}} = \mathbf{K}_{\text{b}}^{\text{Mat}} + \mathbf{K}_{\text{b}}^{\text{Geo}}$ (Lec. 3, slides 10-12).

The **material stiffness** is the classical Euler-Bernoulli stiffness:

$$\mathbf{K}_{\text{b}}^{\text{Mat}} = \frac{EI_z}{L^3}\begin{bmatrix} 12 & 6L & -12 & 6L \\ 6L & 4L^2 & -6L & 2L^2 \\ -12 & -6L & 12 & -6L \\ 6L & 2L^2 & -6L & 4L^2 \end{bmatrix} = 5400\begin{bmatrix} 12 & 6 & -12 & 6 \\ 6 & 4 & -6 & 2 \\ -12 & -6 & 12 & -6 \\ 6 & 2 & -6 & 4 \end{bmatrix} \tag{1.4}$$

The **geometric stiffness** (Lec. 3, slide 12) depends on the current resultant axial force P_0:

$$\mathbf{K}_{\text{b}}^{\text{Geo}} = \frac{P_0}{30L}\begin{bmatrix} 36 & 3L & -36 & 3L \\ 3L & 4L^2 & -3L & -L^2 \\ -36 & -3L & 36 & -3L \\ 3L & -L^2 & -3L & 4L^2 \end{bmatrix} \tag{1.5}$$

Since the classical beam element used here has no axial degrees of freedom, no axial stretch develops and $P_0 = 0$ throughout. Therefore $\mathbf{K}_{\text{b}}^{\text{Geo}} = \mathbf{0}$ at all iterations, and:

$$\mathbf{K}_{\text{b}} = \mathbf{K}_{\text{b}}^{\text{Mat}} \tag{1.6}$$

Applying the clamped boundary conditions ($w_1 = \theta_1 = w_5 = \theta_5 = 0$), the reduced contributions of the two beam elements to Node 2 DOFs $\{w_2, \theta_2\}$ are:

$$\mathbf{K}_{\text{free}}^{(1)} = 5400\begin{bmatrix} 12 & -6 \\ -6 & 4 \end{bmatrix}, \qquad \mathbf{K}_{\text{free}}^{(2)} = 5400\begin{bmatrix} 12 & 6 \\ 6 & 4 \end{bmatrix} \tag{1.7}$$

The assembled beam stiffness in $\{w_2, \theta_2\}$ is:

$$\mathbf{K}_{\text{global}}^{\text{beam}} = \mathbf{K}_{\text{free}}^{(1)} + \mathbf{K}_{\text{free}}^{(2)} = 5400 \begin{bmatrix} 24 & 0 \\ 0 & 8 \end{bmatrix} = \begin{bmatrix} 1.296 \times 10^5 & 0 \\ 0 & 4.32 \times 10^4 \end{bmatrix} \tag{1.8}$$

The off-diagonal terms cancel by symmetry. Consequently θ_2 decouples:

$$4.32 \times 10^4 \, \theta_2 = 0 \implies \theta_2 = 0 \quad \text{at every iteration.} \tag{1.9}$$

1.2 Truss element (Nodes 3-4)

The truss is vertical and only vertical DOFs are active. With $w_4 = -0.01$ m prescribed, define $\Delta w = w_4 - w_3$.

The Total Lagrangian Green-Lagrange strain (Lec. 3, slide 8):

$$E_{11} = \frac{\Delta w}{L} + \frac{1}{2}\left(\frac{\Delta w}{L}\right)^2 \tag{1.10}$$

The 2^{nd} Piola-Kirchhoff stress, axial force, internal force at Node 3, and Total Lagrangian tangent stiffness (Lec. 3, slide 9):

$$S_{11} = E\,E_{11}, \quad P_0 = EA_0\,E_{11}, \quad f_{\text{int},3}^{\text{truss}} = -P_0, \quad K_{33}^{\text{truss}} = \frac{EA_0 + P_0}{L} \tag{1.11}$$

1.3 Contact penalty formulation

The gap between Node 3 (truss) and Node 2 (beam) is:

$$g = w_3 - w_2 \tag{1.12}$$

Contact is active when $g < 0$. Following the penalty method (Lec. 4, slides 6-8), the contact force and contact tangent stiffness in $\{w_2, w_3\}$ are:

$$\mathbf{f}_{\text{int}}^{\text{contact}} = K_{\text{pen}}\, g \begin{bmatrix} -1 \\ 1 \end{bmatrix}, \qquad \mathbf{K}_c = K_{\text{pen}} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} = 7.2 \times 10^8 \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \text{ N/m} \quad (g < 0) \tag{1.13}$$

1.4 Global assembly

Since θ_2 decouples, the effective system has two free DOFs: $\{w_2, w_3\}$. No forces are applied to the free DOFs ($\mathbf{f}_{\text{ext}} = \mathbf{0}$); the prescribed displacement v_4 enters through the truss strain. The residual is:

$$\mathbf{r} = \begin{bmatrix} -1.296 \times 10^5\, w_2 + K_{\text{pen}}\, g \\ P_0 - K_{\text{pen}}\, g \end{bmatrix} \quad (g < 0), \qquad \mathbf{r} = \begin{bmatrix} -1.296 \times 10^5\, w_2 \\ P_0 \end{bmatrix} \quad (g \geq 0) \tag{1.14}$$

The global tangent stiffness matrix, assembled from beam, truss (Eq. 1.11), and contact contributions, is:

$$\mathbf{K}_T = \begin{bmatrix} 1.296 \times 10^5 + K_{\text{pen}} & -K_{\text{pen}} \\ -K_{\text{pen}} & \dfrac{EA_0 + P_0}{L} + K_{\text{pen}} \end{bmatrix} \quad (g < 0) \tag{1.15}$$

and without contact terms when $g \geq 0$.

1.5 Full Newton-Raphson iterations

The Newton-Raphson update at iteration i is:

$$\mathbf{K}_T^{(i-1)}\,\Delta\mathbf{a}^{(i)} = \mathbf{r}^{(i-1)}, \qquad \mathbf{a}^{(i)} = \mathbf{a}^{(i-1)} + \Delta\mathbf{a}^{(i)} \tag{1.16}$$

1.5.1 Initial state $\mathbf{a}^{(0)}$

$$\mathbf{a}^{(0)} = \begin{bmatrix} w_2 \\ w_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \tag{1.17}$$

With $\Delta w^{(0)} = v_4 - 0 = -0.01$ m:

$$E_{11}^{(0)} = -0.01 + \tfrac{1}{2}(0.01)^2 = -9.95 \times 10^{-3} \tag{1.18}$$

$$P_0^{(0)} = 7.2 \times 10^7 \times (-9.95 \times 10^{-3}) = -7.164 \times 10^5 \text{ N} \tag{1.19}$$

Gap: $g^{(0)} = 0 - 0 = 0$. Contact is **not active** ($g = 0$, no penalty force).

$$\mathbf{r}^{(0)} = \begin{bmatrix} 0 \\ -7.164 \times 10^5 \end{bmatrix} \text{ N} \tag{1.20}$$

$$\mathbf{K}_T^{(0)} = \begin{bmatrix} 1.296 \times 10^5 & 0 \\ 0 & \dfrac{EA_0 + P_0^{(0)}}{L} \end{bmatrix} = \begin{bmatrix} 1.296 \times 10^5 & 0 \\ 0 & 7.128 \times 10^7 \end{bmatrix} \text{ N/m} \tag{1.21}$$

1.5.2 Iteration 1

Solving $\mathbf{K}_T^{(0)}\,\Delta\mathbf{a}^{(1)} = \mathbf{r}^{(0)}$ (diagonal system):

$$\Delta w_2^{(1)} = 0, \qquad \Delta w_3^{(1)} = \frac{-7.164 \times 10^5}{7.128 \times 10^7} = -1.005 \times 10^{-2} \text{ m} \tag{1.22}$$

$$\mathbf{a}^{(1)} = \begin{bmatrix} 0 \\ -1.005 \times 10^{-2} \end{bmatrix} \text{ m} \tag{1.23}$$

1.5.3 Iteration 2

Updated Total Lagrangian quantities with $w_3^{(1)} = -1.005 \times 10^{-2}$ m:

$$\Delta w^{(1)} = -0.01 - (-1.005 \times 10^{-2}) = 5.0 \times 10^{-5} \text{ m} \tag{1.24}$$

$$E_{11}^{(1)} \approx 5.0 \times 10^{-5}, \qquad P_0^{(1)} = 7.2 \times 10^7 \times 5.0 \times 10^{-5} = 3600 \text{ N} \tag{1.25}$$

Gap: $g^{(1)} = -1.005 \times 10^{-2} - 0 = -1.005 \times 10^{-2}$ m < 0. Contact **active**.

$$r_{w_2}^{(1)} = 0 + 7.2 \times 10^8 \times (-1.005 \times 10^{-2}) = -7.236 \times 10^6 \text{ N} \tag{1.26}$$

$$r_{w_3}^{(1)} = 3600 - 7.2 \times 10^8 \times (-1.005 \times 10^{-2}) = 7.240 \times 10^6 \text{ N} \tag{1.27}$$

$$\mathbf{K}_T^{(1)} = \begin{bmatrix} 7.2013 \times 10^8 & -7.2 \times 10^8 \\ -7.2 \times 10^8 & 7.9200 \times 10^8 \end{bmatrix} \text{ N/m} \tag{1.28}$$

Determinant: $\det\left(\mathbf{K}_T^{(1)}\right) = (7.2013)(7.9200) \times 10^{16} - (7.2)^2 \times 10^{16} = 5.143 \times 10^{16}\ \text{N}^2/\text{m}^2$.

$$\Delta w_2^{(2)} = \frac{(-7.236 \times 10^6)(7.9200 \times 10^8) - (-7.2 \times 10^8)(7.240 \times 10^6)}{5.143 \times 10^{16}} = -9.980 \times 10^{-3} \text{ m} \tag{1.29}$$

$$\Delta w_3^{(2)} = \frac{(7.2013 \times 10^8)(7.240 \times 10^6) - (-7.2 \times 10^8)(-7.236 \times 10^6)}{5.143 \times 10^{16}} = 6.796 \times 10^{-5} \text{ m} \tag{1.30}$$

$$w_2^{(2)} = 0 + (-9.980 \times 10^{-3}) = -9.980 \times 10^{-3} \text{ m} \tag{1.31}$$

$$w_3^{(2)} = -1.005 \times 10^{-2} + 6.796 \times 10^{-5} = -9.982 \times 10^{-3} \text{ m} \tag{1.32}$$

1.5.4 Iteration 3

Updated Total Lagrangian quantities with $w_3^{(2)} = -9.982 \times 10^{-3}$ m:

$$\Delta w^{(2)} = -0.01 - (-9.982 \times 10^{-3}) = -1.796 \times 10^{-5} \text{ m} \tag{1.33}$$

$$E_{11}^{(2)} \approx -1.796 \times 10^{-5}, \qquad P_0^{(2)} = 7.2 \times 10^7 \times (-1.796 \times 10^{-5}) = -1293.3 \text{ N} \tag{1.34}$$

Gap: $g^{(2)} = -9.982 \times 10^{-3} - (-9.980 \times 10^{-3}) = -1.796 \times 10^{-6}$ m < 0. Contact remains **active**.

$$r_{w_2}^{(2)} = -(1.296 \times 10^5)(-9.980 \times 10^{-3}) + 7.2 \times 10^8(-1.796 \times 10^{-6}) = 1293.0 - 1293.1 \approx 8.9 \times 10^{-10} \text{ N} \tag{1.35}$$

$$r_{w_3}^{(2)} = -1293.3 - 7.2 \times 10^8(-1.796 \times 10^{-6}) = -1293.3 + 1293.1 \approx 0.17 \text{ N} \tag{1.36}$$

The residual is negligible. The third Newton-Raphson correction is $\Delta\mathbf{a}^{(3)} \approx \mathbf{0}$ ($\sim 10^{-9}$ m), confirming convergence.

1.6 Results

The converged displacements and Newton-Raphson convergence history are given in Table 1.

Table 1: Newton-Raphson iteration history for Problem A.

Iteration	w_2 (m)	w_3 (m)	P_0 (N)	$\|\mathbf{r}\|$ (N)
0	0	0	-7.164×10^5	7.164×10^5
1	0	-1.005×10^{-2}	3.600×10^3	1.024×10^7
2	-9.980×10^{-3}	-9.982×10^{-3}	-1.293×10^3	1.7×10^{-1}
3	-9.980×10^{-3}	-9.982×10^{-3}	-1.293×10^3	≈ 0

The converged displacements are:

$$w_2 = -9.980 \times 10^{-3}\ \text{m} \quad \text{(Node 2 deflects downward by 9.980 mm)} \tag{1.37}$$

$$\theta_2 = 0\ \text{rad} \tag{1.38}$$

$$w_3 = -9.982 \times 10^{-3}\ \text{m} \quad \text{(Node 3 moves downward by 9.982 mm)} \tag{1.39}$$

The residual contact gap is $g = -1.796 \times 10^{-6}$ m, the small elastic overclosure inherent to the penalty method. The force P applied at Node 4 is recovered from the converged truss internal force:

$$E_{11}^{\text{conv}} = -1.796 \times 10^{-5} \tag{1.40}$$

$$P_0^{\text{conv}} = EA_0\, E_{11}^{\text{conv}} = -1293.4\ \text{N} \tag{1.41}$$

$$P = -P_0^{\text{conv}} = 1293.4\ \text{N (downward on Node 4)} \tag{1.42}$$

The truss is in slight compression at convergence: the prescribed displacement of Node 4 forces Node 3 marginally below Node 2, compressing the truss by 1.796×10^{-5} m. The large penalty stiffness ensures both nodes move almost together, with the beam deflection driving the contact force.

2 Problem B: Crack propagation in resin-rich region

The problem concerns crack propagation within a resin-rich region that is adhesively bonded between two rigid bodies. The geometry is depicted in Figure1 of the assignment sheet: a resin block of dimensions $L \times 1.5d$ (width $\times$ height, in the x-y plane) is sandwiched between a top rigid body and a bottom rigid body, each of thickness d. The bottom rigid body is fully clamped (encastre) to the ground. The top rigid body is loaded by a prescribed vertical displacement Δ applied at its top-left node, with roller support in the horizontal direction at that same node. The out-of-plane thickness is taken as $1\,\mu$m, and plane-strain conditions are assumed.

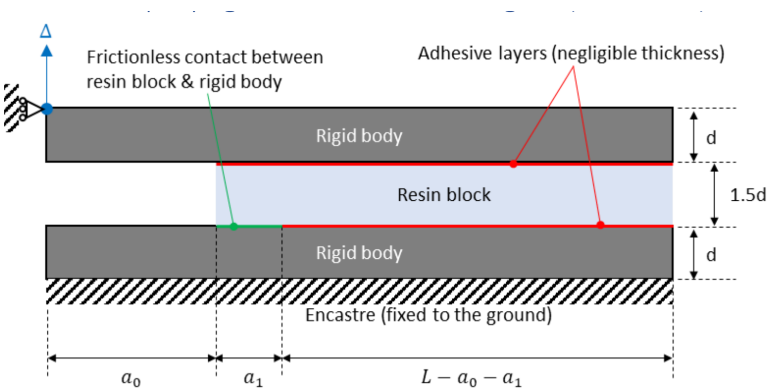

Figure 1: Geometry and boundary conditions for Problem B.

2.1 Finite Element Model Description

2.1.1 Part Definitions, Assembly, and Boundary Conditions

The model consists of three two-dimensional planar parts assembled in Abaqus, as illustrated in Figure 2:

1. **Bottom rigid body** (`Bot`): a rectangular block of dimensions $64\,\mu\text{m} \times 8\,\mu\text{m}$, modelled as a deformable part but constrained as a rigid body via a rigid-body constraint tied to reference point RP-2 located at the top-left corner of the part. An encastre boundary condition is applied to the nodes along the bottom face of this part, fixing all degrees of freedom there.

2. **Resin block** (`Resin`): a deformable block of dimensions $64\,\mu\text{m} \times 12\,\mu\text{m}$, positioned directly above the bottom rigid body. This part constitutes the crack domain. The interface between the resin block and the top rigid body is entirely bonded via surface-based cohesive contact. At the bottom interface, the resin block is partitioned: the right portion (from $x = a_0 + a_1$ to $x = L$) is bonded to the bottom rigid body via cohesive contact, while the left portion (from $x = 0$ to $x = a_0$) has frictionless surface-to-surface contact, representing the pre-existing notch region.

3. **Top rigid body** (`Top`): a rectangular block of dimensions $64\,\mu\text{m} \times 8\,\mu\text{m}$, placed above the resin block. It is constrained as a rigid body tied to reference point RP-1 located at its top-left node. The horizontal displacement of RP-1 is restrained ($u_x = 0$, roller condition), and a smooth-step amplitude ramps the vertical displacement of RP-1 to $\Delta = 10\,\mu$m upward ($+y$) over the full loading step.

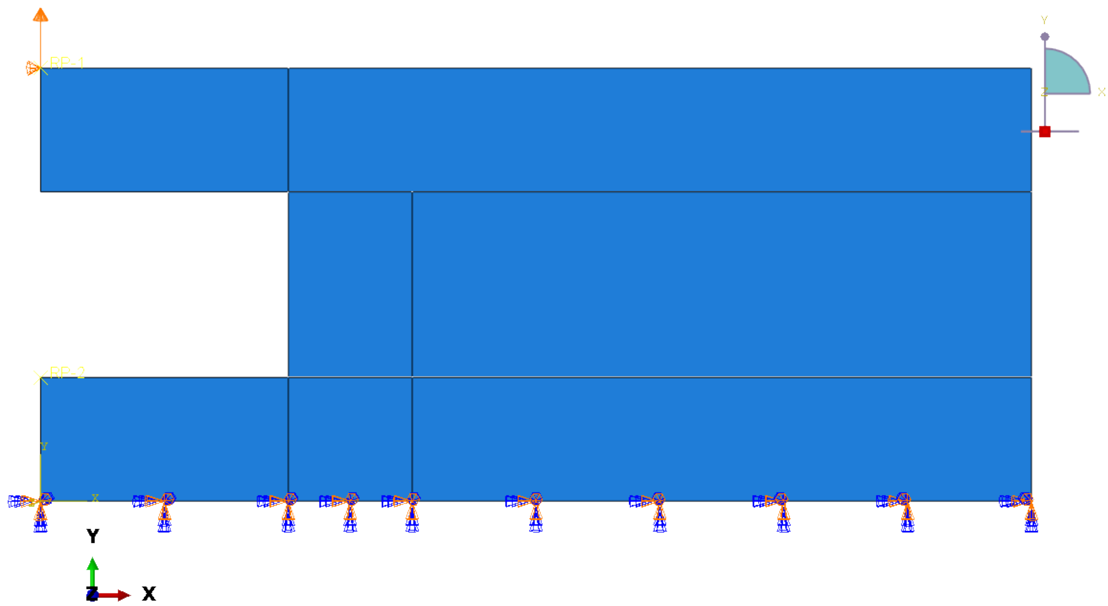

Figure 2: Assembled model with boundary conditions

2.1.2 Mesh

The resin block is discretised using four-node bilinear plane-strain quadrilateral elements with incompatible modes (`CPE4I`). This element type is preferred over the standard reduced-integration formulation because the incompatible mode enrichment alleviates shear locking and improves bending accuracy, which is important in the presence of the stress gradients associated with crack propagation. A uniform mesh of approximately $1\,\mu$m element size is used throughout the resin block to ensure adequate resolution of the crack tip stress field and the cohesive zone along the adhesive interfaces. The top and bottom rigid bodies are discretised using four-node bilinear plane-stress elements with reduced integration (`CPS4R`), at a coarser mesh density.

2.1.3 Crack Modelling - XFEM

Crack initiation and propagation within the resin bulk are modelled using the Extended Finite Element Method (XFEM) with a cohesive-segment approach. The entire resin block is designated as the crack domain. No pre-defined crack path is imposed; the crack propagates along a solution-dependent trajectory determined by the local stress state.

The damage initiation criterion for the resin is the maximum principal stress (MaxPS) criterion:

$$f = \left\langle \frac{\sigma_{\max}}{\sigma^0_{\max}} \right\rangle = 1, \tag{2.1}$$

where $\sigma^0_{\max} = 50\,\text{MPa}$ is the uniaxial tensile strength of the resin and $\langle \cdot \rangle$ denotes the Macaulay bracket, ensuring that crack initiation does not occur under purely compressive stress states. Once the criterion is satisfied, damage evolves according to an energy-based linear cohesive (softening) law with a mode-I fracture toughness of $G_c = 0.15\,\text{kJ/m}^2$.

To improve numerical convergence in the presence of the softening response and the associated snap-back behaviour, viscous regularisation with a stabilisation coefficient of $\mu = 10^{-4}$ is applied to the XFEM damage model.

2.1.4 Adhesive Interface Modelling - Surface Cohesive Behaviour

The interfaces between the resin block and each rigid body are modelled using surface-based cohesive behaviour assigned to contact pairs. Adhesive damage initiation is governed by a quadratic nominal stress (QUADS) criterion:

$$\left(\frac{\langle t_n \rangle}{t_n^0}\right)^2 + \left(\frac{t_s}{t_s^0}\right)^2 = 1, \tag{2.2}$$

where $t_n^0 = 60\,\mathrm{MPa}$ and $t_s^0 = 90\,\mathrm{MPa}$ are the normal and shear interface strengths, respectively.

Damage evolution is governed by an energy-based law with the Benzeggagh-Kenane (BK) mixed-mode criterion:

$$G_c = G_{Ic} + (G_{IIc} - G_{Ic}) \left(\frac{G_{\text{shear}}}{G_T}\right)^\eta, \tag{2.3}$$

where $G_{Ic} = 0.21\,\mathrm{kJ/m^2}$, $G_{IIc} = 0.8\,\mathrm{kJ/m^2}$, $G_{\text{shear}} = G_{II} + G_{III}$, $G_T = G_I + G_{\text{shear}}$, and $\eta = 1.5$. Viscous stabilisation ($\mu = 10^{-4}$) is also applied to the adhesive damage.

Frictionless surface-to-surface contact with a penalty formulation is prescribed between the left portion of the resin block bottom face and the corresponding surface of the bottom rigid body, preventing interpenetration without transmitting cohesive tractions.

2.1.5 Solution Procedure

The analysis is performed as a single quasi-static step in Abaqus with geometric non-linearity enabled. The automatic time-stepping scheme uses an initial increment size of 0.01, a minimum of 10^{-5}, and a maximum of 0.1, over a total step time of 1.0 (corresponding to the full applied displacement of $10\,\mu$m). Up to 5000 increments are permitted, with a maximum of 15 cutbacks per increment. A smooth-step amplitude is used for the displacement boundary condition to avoid artificial dynamic effects during the initial phase of loading.

2.2 Results

2.2.1 Load-Displacement Curve

The predicted load-displacement curve, extracted from the vertical reaction force at RP-1 versus the applied displacement Δ, is presented in Figure 3.

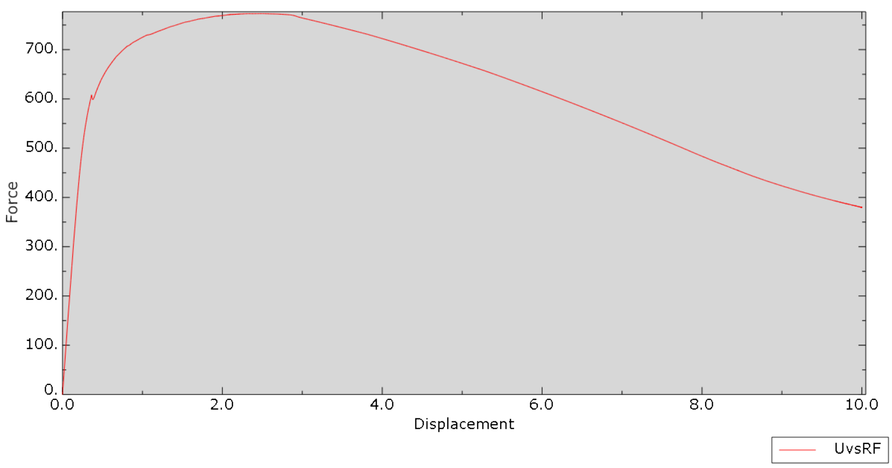

Figure 3: Predicted load-displacement curve up to $\Delta = 10\,\mu$m.

The response exhibits three distinct phases. During the initial phase ($\Delta \lesssim 0.3\,\mu$m), the system behaves approximately linear-elastically as the resin block and adhesive interfaces accumulate load without damage. A visible kink and slight local force drop near $\Delta \approx 0.3\,\mu$m (force $\approx 600\,\mu$N) marks the onset of damage, corresponding to crack initiation at the left notch tip where the maximum principal stress first reaches $\sigma^0_{\max} = 50\,$MPa.

Following this event, the force continues to increase as the crack propagates stably and the remaining bonded ligament carries the load, reaching a global peak of approximately 765 N at $\Delta \approx 2.0\,\mu$m. Beyond this peak, the reaction force decreases monotonically, reflecting progressive crack growth through the resin and continued debonding of the adhesive interfaces. By the end of the analysis at $\Delta = 10\,\mu$m, the force has dropped to approximately 380 N, indicating extensive fracture of the specimen.

2.2.2 Stress Concentration at Crack Onset

Figure 4 shows the von Mises stress distribution at an early increment, corresponding to the onset of crack propagation. The stress concentration is clearly localized at the tip of the left-side notch ($x = a_0 = 16\,\mu$m from the left edge) at the bottom interface of the resin block. This point signifies where the interface between the resin and the bottom rigid body transitions from frictionless contact to cohesive behavior. This confirms that crack initiation is controlled by the stress singularity change in interface conditions, consistent with the application of the MaxPS criterion.

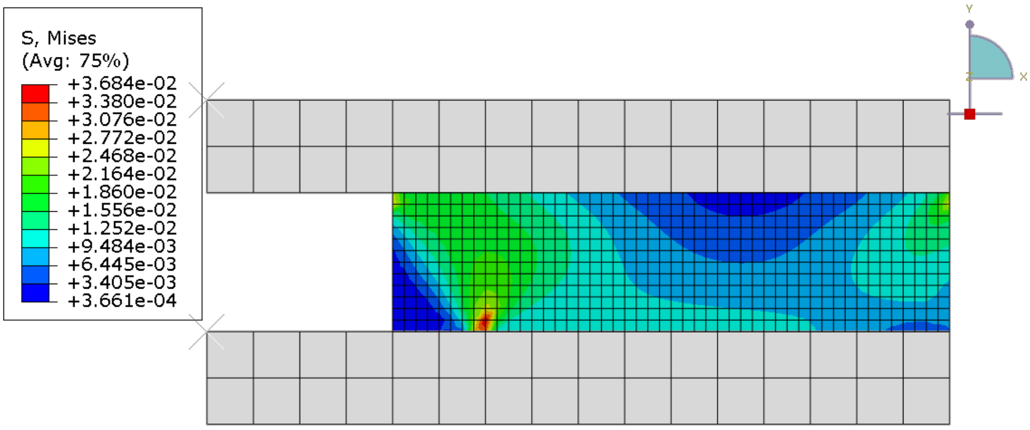

Figure 4: Von Mises stress at crack onset, showing the stress concentration at the left notch tip.

2.2.3 Crack Path

Figure 5 shows the PSILSM (signed-distance function) contour at the final deformation increment ($\Delta = 10\,\mu$m, deformation scale factor of 1). The PSILSM field is used to visualise the XFEM crack surface: elements in which the signed-distance function transitions from negative to positive values are bisected by the crack, and fully damaged elements appear in dark blue.

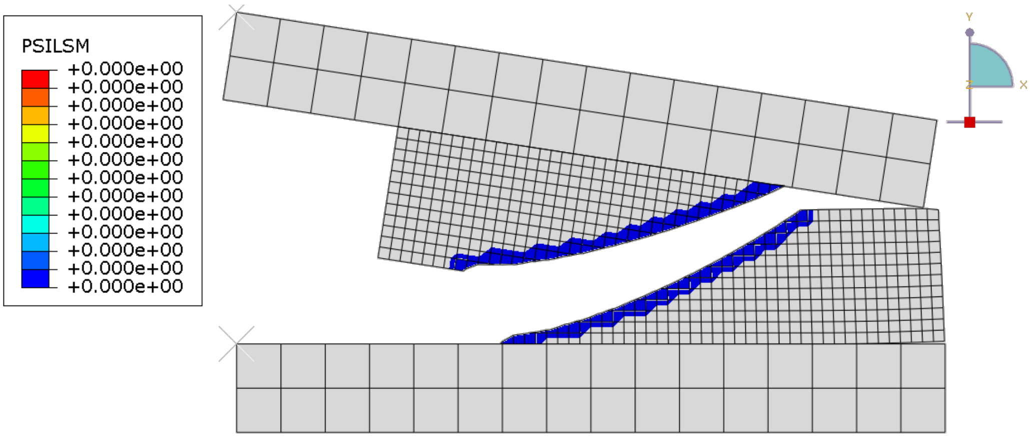

Figure 5: PSILSM contour at $\Delta = 10\,\mu$m (scale factor = 1). Dark blue elements indicate the diagonal XFEM crack path through the resin block.

The XFEM crack initiates at the tip of the left notch at the top interface ($x = 16\,\mu$m) and propagates in a predominantly diagonal direction, traversing the full height of the resin block. This diagonal trajectory is consistent with the mixed-mode (Mode I and Mode II) character of the loading: the eccentric upward displacement applied at the top-left reference point generates both tensile opening and shear sliding across the resin block.

2.2.4 Final Deformed Geometry and Stress Field

Figure 6 shows the von Mises stress field at the final increment ($\Delta = 10\,\mu$m) with a deformation scale factor of 1. The crack faces are fully open and the two portions of the resin block have rotated substantially relative to each other, following the kinematics imposed by the rigid bodies.

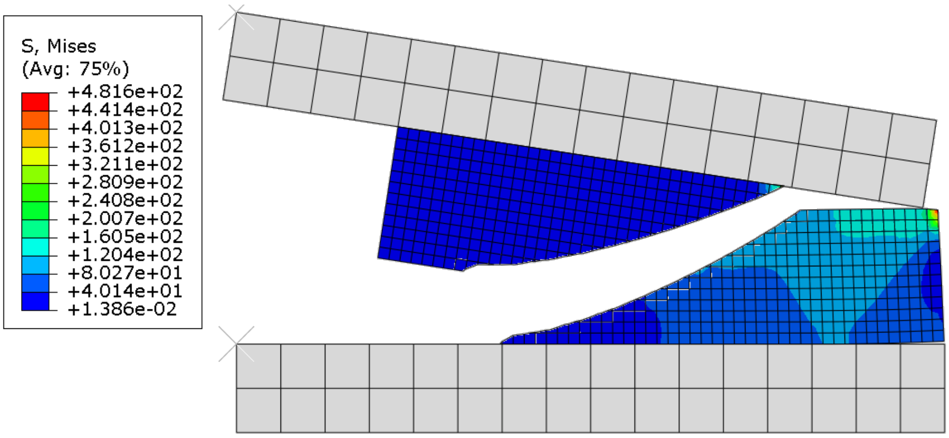

Figure 6: Von Mises stress at the final increment ($\Delta = 10\,\mu$m, scale factor = 1), showing the open crack.

The separated fragments of the resin block experience very low von Mises stresses (dark blue), consistent with the near-complete loss of load-carrying capacity in those regions. The top resin portion, which is fully separated, carries near-zero stress, while the bottom is still in partial contact with the upper rigid body exhibits slightly elevated stresses. The resin fragments retain a permanently deformed shape consistent with the accumulated inelastic displacements during crack propagation.

2.3 Discussion

The maximum principal stress criterion is physically appropriate for brittle resin materials, in which crack initiation is governed by a critical tensile stress rather than a shear or mixed-mode criterion. The crack originates at the tip of the left notch because the geometric discontinuity produces a stress concentration that drives the maximum principal stress to the threshold of 50 MPa at a relatively modest applied displacement. The resulting diagonal crack path demonstrates the mixed-mode character of the loading: the eccentric application of the upward displacement at the top-left node creates a combination of peel (Mode I) and shear (Mode II) loading across the resin cross-section, driving the crack at an angle rather than straight across the block.

The coupled use of XFEM for bulk crack propagation and surface cohesive behaviour for interface debonding enables the simulation to capture the interaction between these two distinct failure modes. As the top rigid body rotates under the applied displacement, the adhesive interface experiences progressive debonding initiating from the edges of the bonded region where peel stresses are highest. This interface debonding reduces the kinematic constraint on the resin block, which in turn promotes further crack advance within the bulk. The interplay between these two mechanisms - bulk fracture and interface delamination - produces the characteristic post-peak load-shedding observed in the load-displacement curve.

The BK mixed-mode criterion is well-suited for the adhesive interface, since the applied loading induces a combination of normal (Mode I) and shear (Mode II) tractions. The BK exponent $\eta = 1.5$ governs the mode-mix dependence of the critical energy release rate; the substantially higher Mode II toughness ($G_{IIc} = 0.8\,\mathrm{kJ/m^2}$) compared to Mode I ($G_{Ic} = 0.21\,\mathrm{kJ/m^2}$) reflects the greater resistance of the adhesive to sliding failure relative to peel failure, which is typical of structural adhesives.

The smooth-step amplitude function ensures that the loading is introduced gradually, minimising spurious dynamic effects in the quasi-static analysis. The viscous stabilisation coefficient ($\mu = 10^{-4}$) is sufficiently small to preserve the accuracy of the global load-displacement response while maintaining numerical convergence through the unstable crack propagation events associated with the softening material behaviour.

References

[1] Dr. Boyang Chen, *Lecture Notes - Non-Linear Modelling (using F.E.M.)*, Brightspace.